

factor. However, there are specific design traits that can determine how comfortable a keyboard is."

Jain adds, "Research indicates that the keyboard height and slope can increase wrist extension, thus increasing the risk of hand, forearm and upper extremity discomfort. As a result, you might want to consider a thin and flat keyboard design."

"The Logitech Cordless Desktop LX and MX family of keyboards, for example, are all 22.3 mm in height. They also feature 'zero degree tilt', which means the primary rows of keys are all at the same height. Also, keyboards that offer navigation controls complementary to those on the mouse also have ergonomic benefits," says Jain.

Indeed, in most ergonomic keyboards, the key mechanisms have become flatter and lighter, and require less force. An ideal ergonomic design should eliminate the inadequacies of the keyboard and mouse. It should be flexible to meet unique user needs, in regard to dimensions and otherwise, of all users.

Ergonomic input devices let you adjust your position more easily and stay further back from the screen, aiding flexibility and easing stress on the eyes, back and shoulders.

Basic Inadequacies

Have you noticed that the rows of alphanumeric keys on the keyboard don't line up in columns? This arrangement was first introduced in the mechanical typewriter to prevent the levers from colliding. Obviously, this design serves no function on an electronic keyboard.

There is no particular justification for the layout of the keys

in the QWERTY format. The Dvorak layout is an alternative to the familiar QWERTY. But despite the advantages of the Dvorak layout, QWERTY has remained the *de facto* standard.

"I don't routinely recommend Dvorak to RSI patients, but encourage them to try out various modifications to see what works for them. At the very least, when they first make the switch, the unfamiliar layout slows them down, and assists the healing process in injured arms and wrists," says Dr Deepak Sharan, a Bangalore-based orthopaedic surgeon, who also provides consultancy in Ergonomics/RSI (and offers rehabilitation) to top IT companies in India.

Dr Sharan feels, "In general, any design that promotes a neutral posture of the wrist is helpful. However, like every other ergonomic tool, different people respond differently to the same product, and it's impossible to make generalised recommendations for everybody."

Preventing Injuries

The Carpal Tunnel Syndrome (CTS) and RSI are caused by a continuous stressful motion of the fingers, wrists, elbows, shoulders, back, and neck. Soreness, numbness, a tingling feeling and wrist pain are other symptoms.

"Over 75 per cent of IT professionals in India report musculoskeletal symptoms of varying severity at work," says Dr Sharan. Srikant Makarani, underwent myotherapy



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Orthopaedic consultant



The Logitech V500 Cordless Mouse has an expandable chassis

treatment to heal a painful wrist condition. Makarani feels that had he used an ergonomic keyboard and mouse at the outset, the problem would not have been so severe.

Makarani, has now switched to the Microsoft Natural Keyboard Elite series, and vouches for the comfort they offer over conventional keyboards.

Neeru Pahwa has been in the IT industry for the past seven years. An engineer with an MNC, she is on leave to recuperate from RSI. Pahwa says, "A year ago, I developed a painful sensation in my wrists. At first, most doctors could not correctly diagnose the reason for the injury. It was very frustrating. In a few months, the pain extended to my shoulders and neck. And now, it has affected and worsened the nerves as well."

At present, Pahwa is undergoing daily treatment for her ailment. She strongly feels that increased awareness about ergonomic devices coupled with information on correct posture will spare others from facing a similar predicament.

Preliminary findings from Dr Sharan's four-year study on RSI have revealed the following predisposing ergonomic factors responsible for computer-related injuries (percentage-wise):

Lack of appropriate breaks (86), improper monitor height (60), mouse being too high (54), resting the arm or wrist on a hard surface while typing (42), keyboard being too high (40), and bizarre leg positioning (25).

Safety And Health

Dr Sharan feels that ergonomics primarily play a preventive role—an important role, but one that cannot compensate for inefficient management or ineffective practical training.

He says, "While inadequacies in the ergonomic design of keyboard and mice do not seem to be major factors causing RSI, we do encourage our patients to try out the various input devices available and see which one works for them."

He goes on to add, "One fundamental problem with the design of alternative keyboards and mice is that it initially feels good because the stress shifts to parts that aren't injured yet. However, it does not address



Tips For A Balanced Computing Posture

- ❑ Balance your head above your torso: don't roll your shoulders or neck forward
- ❑ Place your monitor at a comfortable reading distance, just below eye level, tilted up a little, and free from glare
- ❑ Place your keyboard and mouse on a tray so you don't need to reach up to them
- ❑ Keep your keyboard lowered and angled slightly away (negative pitch)
- ❑ Keep your elbows close by your side: don't reach out. The elbow should be at an angle of 100 to 110 degrees
- ❑ Don't bend your wrists up or out
- ❑ Don't rest your elbows or wrists on anything while typing - no armrests or wrist rests.
- ❑ The feet should be placed firmly on the floor. Otherwise, use a footrest
- ❑ The back should be supported against the chair's backrest
- ❑ Take a micro break: five seconds every five to seven minutes
- ❑ Take a macro break: two minutes every 30 to 40 minutes

Courtesy: Dr Deepak Sharan